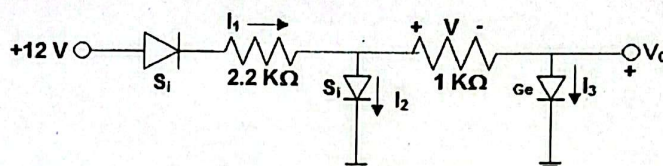


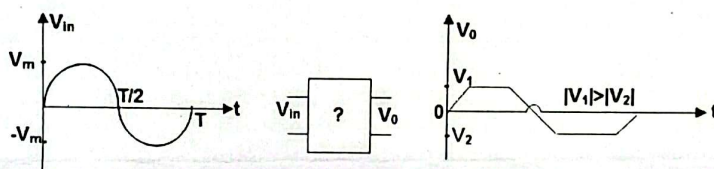
- N.B. (i) Answer any SIX questions taking any THREE from each section.  
(ii) Figures in the right margin indicate full marks.  
(iii) Use separate answer script for each section.

**SECTION : A**

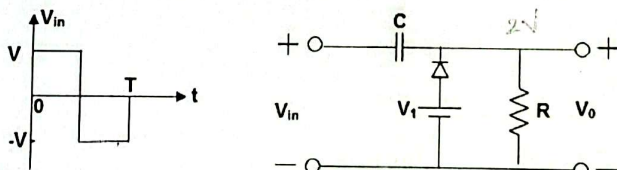
|     |                                                                                                                           | Marks | CO  |
|-----|---------------------------------------------------------------------------------------------------------------------------|-------|-----|
| Q.1 | (a) Write down the values of firing potential for Ge and Si diodes.                                                       | 3     | CO1 |
|     | (b) Starting from the diode equation, derive the expression for dynamic resistance of a p-n junction at room temperature. | 4     | CO1 |
|     | (c) For the network shown below, determine the unknown voltages and currents.                                             | 3     | CO2 |



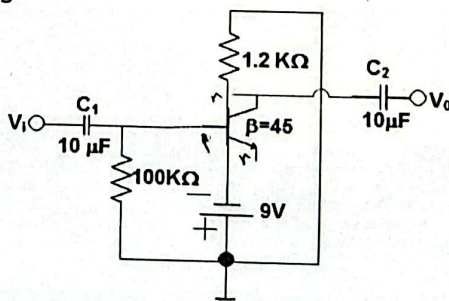
|     |                                                                                                     |   |     |
|-----|-----------------------------------------------------------------------------------------------------|---|-----|
| Q.2 | (a) Show that dc voltage of a full wave rectifier is 0.636 times of peak value of the input signal. | 2 | CO1 |
|     | (b) Design a clipper circuit to obtain the following input-output relationships.                    | 4 | CO2 |



|     |                                            |   |     |
|-----|--------------------------------------------|---|-----|
| (c) | Determine $V_0$ for the following network. | 4 | CO2 |
|-----|--------------------------------------------|---|-----|

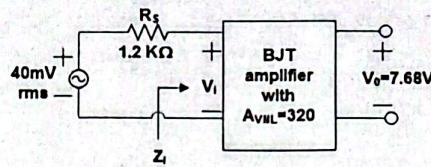


|     |                                                                                                                                                                                                                                                                       |   |     |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|
| Q.3 | (a) Derive the relationship among three current gains for three types (CB, CE, CC) of BJT configurations.                                                                                                                                                             | 2 | CO1 |
|     | (b) Draw the output characteristics of a common-emitter configuration BJT circuit and state how you can get the quiescent point from the output characteristic curves. Also state the desired location of this point in the curve. Explain reason behind your answer. | 4 | CO1 |
|     | (c) For the following BJT network, determine the value of collector voltage and base voltage. State in which region of operation the BJT in the network is working.                                                                                                   | 4 | CO2 |



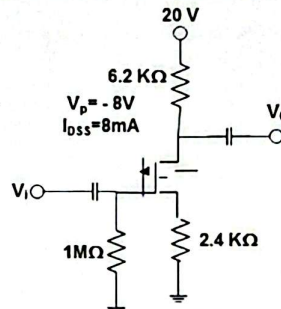
|     |                                                                                                                                                                                                                                                                                               |   |     |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|
| Q.4 | (a) Proof that BJT can be functioned as inverter.                                                                                                                                                                                                                                             | 2 | CO1 |
|     | (b) For a common-emitter configuration BJT amplifier the emitter current, $I_E = 3.2mA$ and its output resistance, $r_o = \infty$ . Determine its input impedance, $Z_i$ and voltage gain, $A_v$ if a $2K\Omega$ load is connected to its output. it is known that $\beta$ of the BJT is 120. | 4 | CO2 |
|     | (c) Calculate the current gain, $A_i$ for the following BJT amplifier if a load of                                                                                                                                                                                                            | 4 | CO2 |

4.8KΩ is connected in the output.

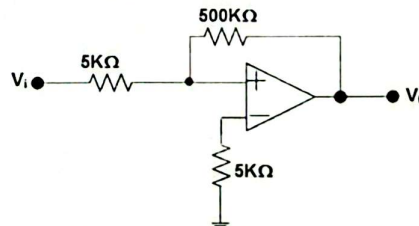


### SECTION : B

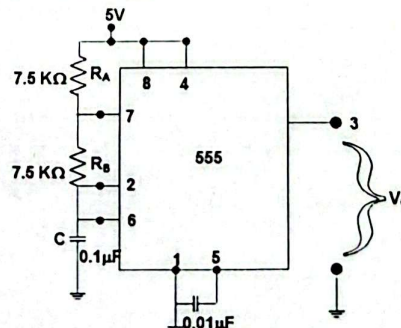
- Q.5 (a) Show that,  $g_m = g_{m0} [1 - \frac{V_{GS}}{V_P}]$ , where the symbols have their usual meanings. 3 CO1
- (b) Sketch the construction of E-type MOSFET and explain the operation along with necessary equations. 3 CO1
- (c) Determine (i)  $I_{DQ}$ , (ii)  $V_{GSQ}$ , (iii)  $V_D$  for the following network. 4 CO2



- Q.6 (a) Design and draw the circuit diagram to fulfill the following mathematical operation where  $V_0$  is the output and  $V_1, V_2$  are inputs.  $V_0 = 6V_1 + 4 \int V_2 dt$  3 CO2
- (b) Define "Slew rate" of an Op-Amp. Writing necessary equation, state how this term is related to the maximum useable frequency of an Op-Amp. 3 CO3
- (c) The following circuit uses an Op-Amp with specified values of input offset voltage,  $V_{IO} = 4mV$  and input offset current of  $I_{IO} = 150 nA$ . Calculate the total offset voltage for the circuit. 4 CO2



- Q.7 (a) Draw the sample and hold circuit, hence, explain the operation with I/O waveform. 3 CO1
- (b) With necessary sketch, explain the operation of 555 timer for monostable multivibrator. 4 CO1
- (c) Draw the voltage across the  $0.1\mu F$  capacitor, c and the output voltage,  $V_0$  for the circuit shown below. Also calculate the frequency and the duty cycle of the output voltage. 3 CO2



- Q.8 (a) Design a low pass first order Butterworth filter at a cut off frequency of 1KHz with a pass band gain of 2. 3 CO2
- (b) Draw the diagram of a DC power supply whose input is an AC. Explain how the circuit works. 4 CO3
- (c) Draw the two transistor model of SCR. State how it works and differ from BJT/FET devices? 3 CO1

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**RAJSHAH UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**
**1<sup>st</sup> Year Even Semester Examination 2023**

COURSE NO: EEE 1251

COURSE TITLE: Electronic Devices and Circuits

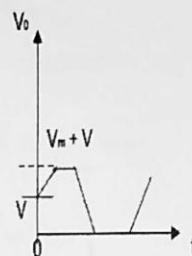
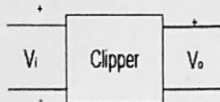
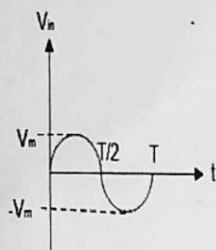
FULL MARKS: 60

TIME: 3 HOURS

- N.B.** (i) Answer any **SIX** questions taking any **THREE** from each section.  
 (ii) Figures in the right margin indicate full marks.  
 (iii) Use separate answer script for each section.  
 (iv) Course Outcomes (COs)

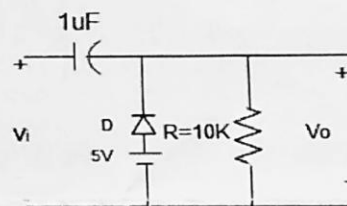
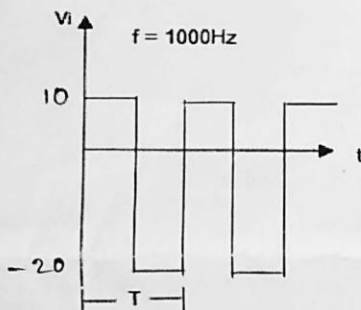
**SECTION : A**

|      |                                                                                                                      | Marks | COs |
|------|----------------------------------------------------------------------------------------------------------------------|-------|-----|
| Q.1. | (a) Compare the characteristics of Si, GaAs, Ge diodes commercial diodes.                                            | 03    | CO1 |
|      | (b) Explain the construction and operation of diode-based AND and OR gates.                                          | 03    | CO1 |
|      | (c) Explain the operation of a center-tapped transformer full-wave rectifier. Also determine the PIV for each diode. | 04    | CO1 |
| Q.2. | (a) Design a clipper circuit to obtain the following input-output relationships.                                     | 04    | CO2 |



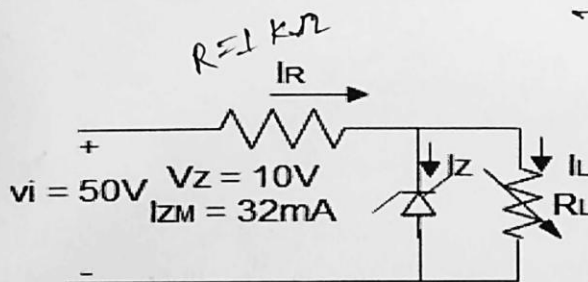
- (b) Determine  $V_0$  for the following network.

03 CO2



- (c) For the following network, determine the range of  $R_L$ ,  $I_L$  that will maintain  $V_{R_L} = 10V$ .

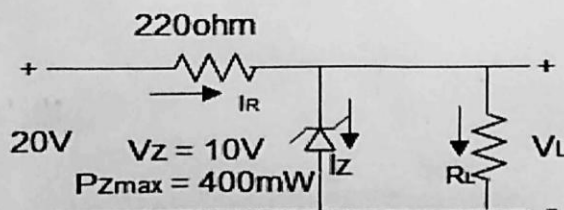
03 CO2



- Q.3. (a) What are the differences between Zener diode and rectifier diode? Indicate the special feature that is available in Zener diode but not in rectifier diode. Write some applications of that feature.
- (b) In the following circuit, determine the minimum value of the load resistance,  $R_L$  to ensure that the Zener diode is in "ON" state.

03 CO1

04 CO2



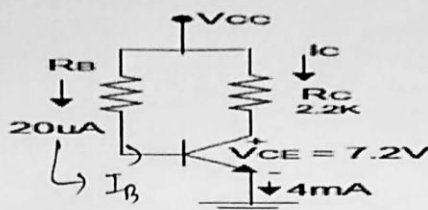
- (c) Draw a full-wave bridge rectifier and explain the circuit's operation with sketching necessary wave shapes.

03 CO1

- Q.4. (a) State the necessary operating regions of a BJT to work if as a switch and as an amplifier.  
 (b) Given the information appearing in the figure below, determine (i)  $I_C$  (ii)  $V_{CE}$  (iii)  $\beta$  and (iv)  $R_B$ .

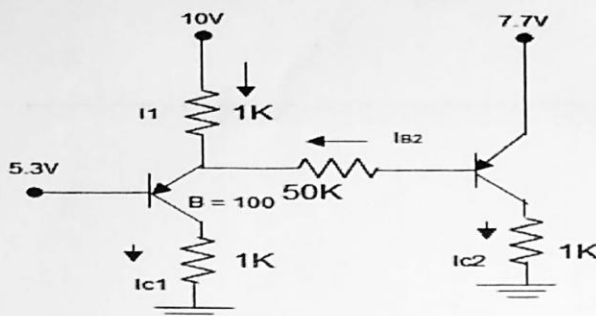
02 CO3

04 CO2



- (c) Find the collector voltages of the two BJTs in the circuit of the below figure.

04 CO2



$$I_E = I_C + I_B$$

$$= I_C + \frac{I_C}{\beta}$$

### SECTION : B

- Q.5. (a) Compare an ideal OP-Amp with  $\mu A741C$  OP-Amp.  
 (b) Explain how slew rate determine the maximum frequency of the input signal in an Op-Amp.  
 (c) Consider an inverting Op-Amp  $\mu A741C$ , with  $R_1 = 100k\Omega$  and  $R_F = 1000k\Omega$ . The input signal is  $V_{in} = 10\sin(300 \times 10^3 t)$  mV. Find the maximum input frequency and comment whether you will get distorted or undistorted output.
- Q.6. (a) Explain the application of OP-Amp as (i) integrator (ii) differentiator (iii) active low pass filter and (iv) current controlled current source.  
 (b) Determine the output voltage of Op-Amp for input voltages of  $V_{i1} = 150 \mu V$ ,  $V_{i2} = 140 \mu V$ ; differential gain  $A_d = 4000$ . Consider the value of CMRR = 100 and  $10^5$ . Comments on your result.  
 (c) Consider the following transfer characteristics and design a Schmitt trigger.

03 CO1

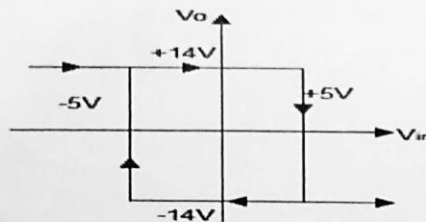
03 CO2

04 CO2

04 CO1

03 CO2

03 CO2



$$A_d = \frac{V_o}{V_d}$$

$$V_o = A_c V_c + A_d V_d$$

$$= A_d V_d \left( 1 + \frac{A_c}{A_d} \cdot \frac{V_c}{V_d} \right)$$

- Q.7. (a) Design a monostable multivibrator using 555 timer so that  $t_p = 5ms$  and  $V_{cc} = 12V$ .  
 (b) Explain the operation of 555 timer as frequency divisor.  
 (c) Explain the application of an UJT as relaxation oscillator.
- Q.8. (a) What do you understand by the term 'virtual ground' in an Op-Amp circuit? Realize the equation of  $V_o = -2V_1 - 3V_2 - V_3$  using Op-Amp.  
 (b) Find the value of the feedback resistor,  $R_F$  in the following circuit to obtain a gain of 5.

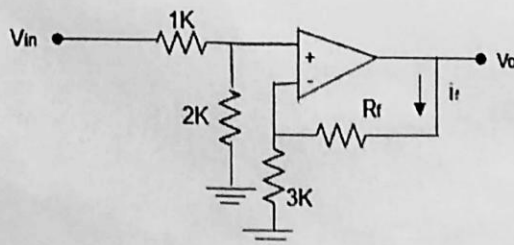
04 CO2

03 CO1

03 CO1

03 CO3

04 CO2



$$19.5 k\Omega$$

- Q.9. What do you understand by class A, class B and class C amplifier?

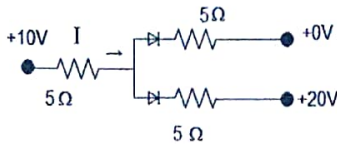
03 CO2

- N.B.
- Answer any SIX questions taking any THREE from each section.
  - Figures in the right margin indicate full marks.
  - Use separate answer script for each section
  - COs-Course Outcomes.

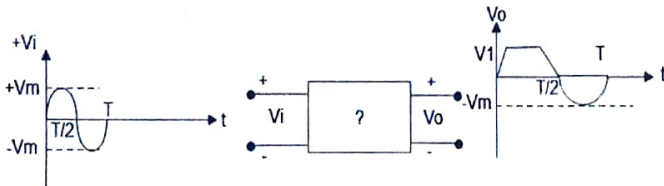
**SECTION A**

COs Marks

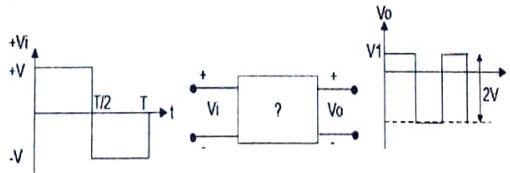
- Q.1. (a) What do you understand by intrinsic and extrinsic semiconductors? Explain those with necessary diagrams and figures. CO1 04
- (b) Explain with necessary sketching how PN-junction is formed. With the biasing arrangement state reverse saturation current. Also indicate the level of this current in the characteristic curve ( $I - V$  characteristics) of a PN-junction diode. CO1 04
- (c) Determine the current  $I$  for the network shown below. The diodes are ideal. CO2 02



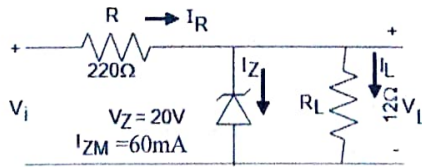
- Q.2. (a) Design a clipper circuit to obtain the following input-output relationships. CO2 03



- (b) Design a clamper circuit to obtain the following input-output relationships. CO2 03



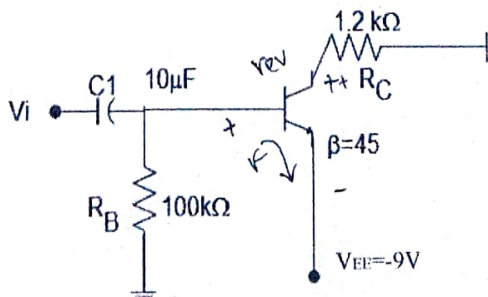
- (c) Determine the range of  $V_i$  that will maintain the Zener diode in the ON state. CO2 04



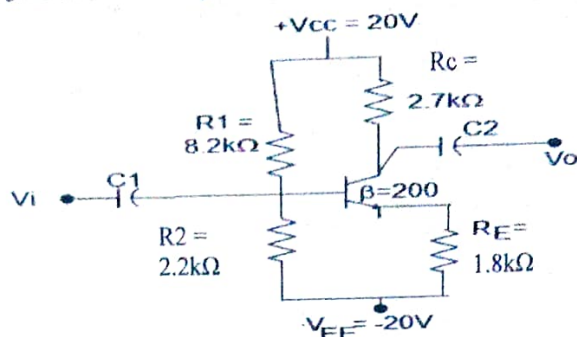
$$V_L = V_Z = \frac{R_L}{R + R_L} V_i$$

$$I_{R_{max}} = I_{ZM}$$

- Q.3. (a) Determine  $V_C$  and  $V_B$  for the following network. CO2 03



- (b) Find  $I_C$  and  $V_C$  for the following network.

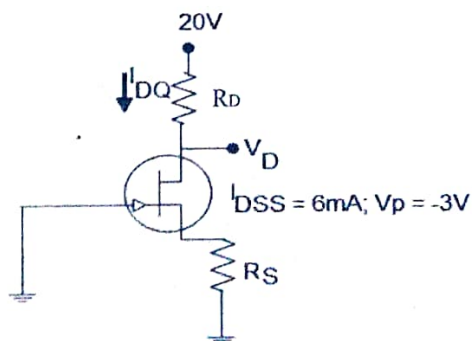


- (c) What is bias stabilization? Which bias is more stable for BJT? Explain in details. CO1 03
- Q.4. (a) Explain the operation of a zero crossing detector. CO1 03
- (b) What is Schmitt trigger? Design a Schmitt trigger so that  $V_{th} = |+V_{th}| = |-V_{th}| = 5V$ . Assume  $V_{sat} = |-V_{sat}| = 14V$ . Draw the transfer characteristics of that circuit. CO2 04
- (c) Draw circuit diagram and explain the operation of active filter circuits as follows: (i) Active band pass filter and (ii) Active stop band filter. Also draw their frequency response curves. CO1 03



### SECTION B

- Q.5. (a) Draw internal block diagram of a 555 timer and explain the function of all pins of this IC. CO1 03
- (b) Explain the operation of a 555 timer as monostable multivibrator. How can this monostable multivibrator be used as pulse wider? CO2 04
- (c) Explain the operation of a 555 timer as a square wave generator. CO2 03
- Q.6. (a) Explain with figure and mathematics how an operational amplifier can be used as an integrator. CO3 04
- (b) What is slew rate of an OP-Amp? Calculate the maximum closed loop voltage gain of an OP-Amp whose slew rate is  $2V/\mu s$  while the input signal varies by 0.5 in 10ns. CO2 04
- (c) Define CMRR of an OP-Amp. What is the expected value of this parameter? CO1 02
- Q.7. (a) What are basic differences between BJT and JFET? Draw the transfer characteristics and drain characteristics of an N-channel JFET and explain those in brief. CO1 04
- (b) For the following network determine the value of  $R_D$  and  $R_S$  for  $I_{DQ} = 2.5mA$  and  $V_D = 12V$ . Calculate also the maximum power handled by the FET at this operating condition. CO3 04



$$\frac{0.5}{10 \times 10^{-9}} \times 0.01 \times 10^{-6}$$

- (c) Draw the  $r_e$  model for the common-emitter transistor configuration and common-source FET ac equivalent model. CO2 02
- Q.8. (a) Show that the maximum efficiency of a class A amplifier is 50%. CO4 04
- (b) Draw the two transistor equivalent model of an SCR. State how it works and what special characteristics we can have from this device in contrast to BJT and FET. CO1 04
- (c) What is TRIAC? State an application of this device. CO1 02

Heaven's Light Is Our Guide  
**RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

1<sup>st</sup> Year Even Semester Examination 2021

COURSE NO: **EEE 1251**      COURSE TITLE: **Electronic Devices and Circuits**

FULL MARKS: 60

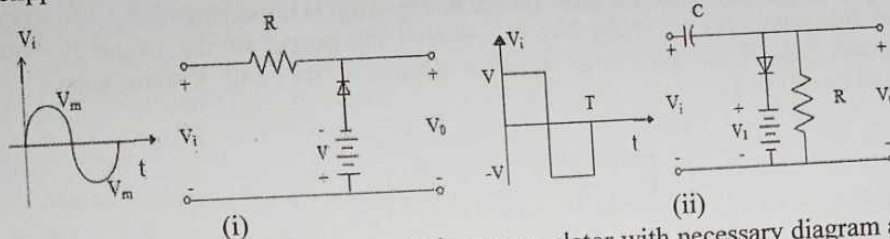
TIME: 3 HRS

- N.B. (i) Answer any **SIX** questions taking any **THREE** from each section.  
(ii) Figures in the right margin indicate full marks.  
(iii) Use separate answer script for each section.

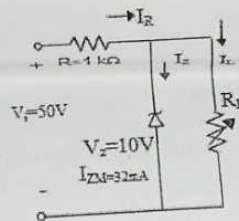
**SECTION : A**

COs    Marks

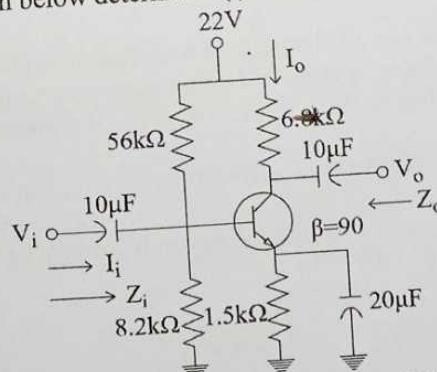
- Q.1. (a) What is reverse recovery time. Briefly explain the comparative I-V characteristics curve for *Ge*, *Si* and *GaAs* semiconductor. CO1    3  
(b) Explain the operation of full wave bridge rectifier circuit with *Si* diodes and hence find its output DC value and ripple factor. CO1    3  
(c) Define clipper and clamper circuit. Find the output wave shape of the following clipper and clamper circuit with necessary explanation. CO1    4



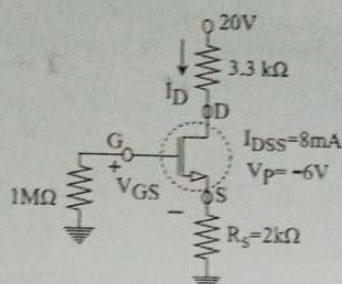
- Q.2. (a) Explain how you can use a zener diode as a regulator with necessary diagram and numerical values. CO1    3  
(b) For the following figure determine the range of  $R_L$  and  $I_L$  that will result in  $V_{RL}$  being maintained at 10V. Also determine the maximum wattage rating of the diode. CO1    3



- (c) Build a BJT switching circuit and correspondingly explain how BJT works as switch. CO2    4  
Q.3. (a) Briefly explain the construction and operation of a p-n-p bipolar junction transistor. CO2    4  
(b) For the network shown below determine- (i)  $Z_i$ , (ii)  $Z_o$ , and (iii)  $A_v$  if  $\pi_0 = 50 \text{ k}\Omega$ . CO2    3

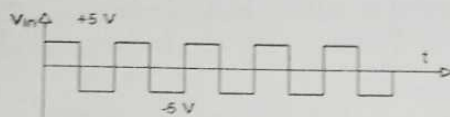


- (c) Explain how you can design AND, OR logic gate using BJT. CO2    3  
Q.4. (a) Briefly explain the operation of a depletion type NMOS with necessary figures and characteristics. CO3    4  
(b) What is CMOS? Design an inverter circuit using CMOS and explain how it works as an inverter. CO3    3  
(c) Determine the following for the network shown below:  
(i)  $V_{GSQ}$ , (ii)  $I_{DQ}$ , (iii)  $V_{DS}$  (iv)  $V_S$ , (v)  $V_G$  (vi)  $V_D$  CO3    3

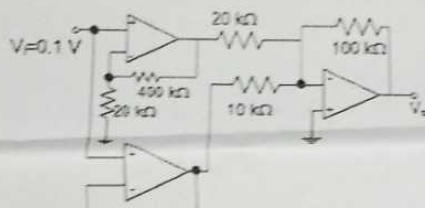


### SECTION : B

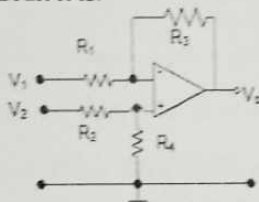
- Q.5. (a) What are the characteristic features of an ideal Op-Amp (IC 741)? Write some applications of this IC. CO4 3
- (b) Draw a unity follower circuit using 741 IC. Why do we need unity follower? CO4 2
- (c) What is the desired value of an Op-Amp? An amplifier by using Op-Amp has a differential gain of 4000 and the value of CMRR is  $10^5$ . Determine the output voltage of that amplifier for the input voltages of  $V_{i1}=150 \mu V$  and  $V_{i2}=140 \mu V$ . CO4 3
- (d) As an engineer why do you think slew rate of an Op-Amp is important? CO4 2
- Q.6. (a) Draw an integrator circuit using 741 IC. Sketch the output of the circuit in approximate scale if the following signal is applied at its input. Choose time constant of circuit as 1. CO4 4



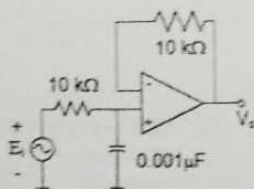
- (b) Calculate the output voltage for the circuit shown below. CO4 3



- (c) Derive the expression of the output of the following circuit choosing  $R_1=R_2$  and  $R_3=R_4$ . Write what type of circuit it is. CO4 3



- Q.7. (a) using 555 time IC draw a circuit diagram of a monostable multivibrator circuit and explain its operation drawing input and output curves from the circuit. CO4 3
- (b) What do you mean by duty cycle of a multivibrator? Explain with necessary diagram. CO4 2
- (c) Using Op-Amp construct a band pass active filter circuit. State how the circuit works. CO5 2
- (d) Calculate the cutoff frequency of the circuit shown below. What type of circuit is it? What is the gain of the circuit? CO5 3



- Q.8. (a) Draw the two transistor model of SCR. State how it works and differ from 3 layers devices like BJT, FET. CO5 3
- (b) What is TRIAC? Draw a circuit diagram using TRIAC that can control the intensity of light (Light Dimmer Circuit) and explain its operation. CO5 4
- (c) Draw the diagram of a regulated DC power supply using semiconductor devices, and explain how it works. CO5 3